

SOFTWARE

- TriangleDB
- TrickBot
- TrickMo
- Triton
- Trojan-SMS.AndroidOS.Agent.ao
- Trojan-SMS.AndroidOS.FakeInst.a
- Trojan-SMS.AndroidOS.OpFake.a
- Trojan.Karagany
- Trojan.Mebromi
- Troll Stealer
- TruffleHog
- Truvasys
- TSCookie
- Tsundere Botnet
- Turian
- TURNEDUP
- Twitoor
- TYPEFRAME
- UACMe
- UBoatRAT
- Umbreon
- Unknown Logger

Home > Software > TrickBot

TrickBot

TrickBot is a Trojan spyware program written in C++ that first emerged in September 2016 as a possible successor to Dyre. TrickBot was developed and initially used by Wizard Spider for targeting banking sites in North America, Australia, and throughout Europe; it has since been used against all sectors worldwide as part of “big game hunting” ransomware campaigns.

ID: S0266

Associated Software: Totbrick, TSPY_TRICKLOAD

Type: MALWARE

Platforms: Windows

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Version: 2.2

Created: 17 October 2018

Last Modified: 10 April 2024

Version Permalink

Associated Software Descriptions

Name	Description
Totbrick	[5] [6]
TSPY_TRICKLOAD	[5]

Techniques Used

ATT&CK® Navigator Layers -

Domain	ID	Name	Use
Enterprise	T1087	.001 Account Discovery: Local Account	TrickBot collects the users of the system.[1][7]
		.003 Account Discovery: Email Account	TrickBot collects email addresses from Outlook.[7]
Enterprise	T1071	.001 Application Layer Protocol: Web Protocols	TrickBot uses HTTPS to communicate with its C2 servers, to get malware updates, modules that perform most of the malware logic and various configuration files.[1][8]
Enterprise	T1547	.001 Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	TrickBot establishes persistence in the Startup folder.[9]
Enterprise	T1185	Browser Session Hijacking	TrickBot uses web injects and browser redirection to trick the user into providing their login credentials on a fake or modified web page.[2][3][6][7]
Enterprise	T1110	.004 Brute Force: Credential Stuffing	TrickBot uses brute-force attack against RDP with rdpscanDll module.[9][10]
Enterprise	T1059	.001 Command and Scripting Interpreter: PowerShell	TrickBot has been known to use PowerShell to download new payloads, open documents, and upload data to command and control servers.[11]
		.003 Command and Scripting Interpreter: Windows Command Shell	TrickBot has used macros in Excel documents to download and deploy the malware on the user's machine.[12]
Enterprise	T1543	.003 Create or Modify System Process: Windows Service	TrickBot establishes persistence by creating an autostart service that allows it to run whenever the machine boots.[7]
Enterprise	T1555	.003 Credentials from Password Stores: Credentials from Web Browsers	TrickBot can obtain passwords stored in files from web browsers such as Chrome, Firefox, Internet Explorer, and Microsoft Edge, sometimes using esentutil.[7][8][11]
		.005 Credentials from Password Stores: Password Managers	TrickBot can steal passwords from the KeePass open source password manager.[8]
Enterprise	T1132	.001 Data Encoding: Standard Encoding	TrickBot can Base64-encode C2 commands.[8]
Enterprise	T1005	Data from Local System	TrickBot collects local files and information from the victim's local machine.[11]
Enterprise	T1140	Deobfuscate/Decode Files or Information	TrickBot decodes the configuration data and modules.[2][8][13]
Enterprise	T1685	Disable or Modify Tools	TrickBot can disable Windows Defender.[7]
Enterprise	T1482	Domain Trust Discovery	TrickBot can gather information about domain trusts by utilizing Nttest.[14][8]
Enterprise	T1573	.001 Encrypted Channel: Symmetric Cryptography	TrickBot uses a custom crypter leveraging Microsoft's CryptoAPI to encrypt C2 traffic.[2]Newer versions of TrickBot have been known to use bcrypt to encrypt and digitally sign responses to their C2 server.[15]
Enterprise	T1041	Exfiltration Over C2 Channel	TrickBot can send information about the compromised host and upload data to a hardcoded C2 server.[8][11]
Enterprise	T1210	Exploitation of Remote Services	TrickBot utilizes EternalBlue and EternalRomance exploits for lateral movement in the modules wormwinDll, wormDll, mwormDll, nwormDll, tabDll.[9]
Enterprise	T1008	Fallback Channels	TrickBot can use secondary C2 servers for communication after establishing connectivity and relaying victim information to primary C2 servers.[8]
Enterprise	T1083	File and Directory Discovery	TrickBot searches the system for all of the following file extensions: .avi, .mov, .mkv, .mpeg, .mpeg4, .mp4, .mp3, .wav, .ogg, .jpeg, .jpg, .png, .bmp, .gif, .tiff, .ico, .xlsx, and .zip. It can also obtain browsing history, cookies, and plug-in information.[1][7]
Enterprise	T1495	Firmware Corruption	TrickBot module “Trickboot” can write or erase the UEFI/BIOS firmware of a compromised device.[16]
Enterprise	T1564	.003 Hide Artifacts: Hidden Window	TrickBot has used a hidden VNC (hVNC) window to monitor the victim and collect information stealthily.[17]
Enterprise	T1105	Ingress Tool Transfer	TrickBot downloads several additional files and saves them to the victim's machine.[9][11]
Enterprise	T1056	.004 Input Capture: Credential API Hooking	TrickBot has the ability to capture RDP credentials by capturing the CredEnumerateA API[12]
Enterprise	T1559	.001 Inter-Process Communication: Component Object Model	TrickBot used COM to setup scheduled task for persistence.[9]
Enterprise	T1036	Masquerading	The TrickBot downloader has used an icon to appear as a Microsoft Word document.[8]
Enterprise	T1498	Modify System	TrickBot can modify system registry.[7]

Enterprise	T1114		module registry	TrickBot can modify registry entries. ^[1]
Enterprise	T1106		Native API	TrickBot uses the Windows API call, CreateProcessW(), to manage execution flow. ^[1] TrickBot has also used <code>nt!*</code> API functions to perform Process Injection. ^[13]
Enterprise	T1135		Network Share Discovery	TrickBot module shareDll/mshareDll discovers network shares via the WNetOpenEnumA API. ^{[9][10]}
Enterprise	T1571		Non-Standard Port	Some TrickBot samples have used HTTP over ports 447 and 8082 for C2. ^{[1][2][5]} Newer versions of TrickBot have been known to use a custom communication protocol which sends the data unencrypted over port 443. ^[11]
Enterprise	T1027		Obfuscated Files or Information	TrickBot uses non-descriptive names to hide functionality. ^[1]
		.002	Software Packing	TrickBot leverages a custom packer to obfuscate its functionality. ^[1]
		.013	Encrypted/Encoded File	TrickBot uses an AES CBC (256 bits) encryption algorithm for its loader and configuration files. ^[1]
Enterprise	T1069		Permission Groups Discovery	TrickBot can identify the groups the user on a compromised host belongs to. ^[8]
Enterprise	T1566	.001	Phishing: Spearphishing Attachment	TrickBot has used an email with an Excel sheet containing a malicious macro to deploy the malware. ^[12]
		.002	Phishing: Spearphishing Link	TrickBot has been delivered via malicious links in phishing e-mails. ^[8]
Enterprise	T1542	.003	Pre-OS Boot: Bootkit	TrickBot can implant malicious code into a compromised device's firmware. ^[14]
Enterprise	T1057		Process Discovery	TrickBot uses module networkDll for process list discovery. ^{[9][10]}
Enterprise	T1055		Process Injection	TrickBot has used <code>nt!*</code> Native API functions to inject code into legitimate processes such as <code>services.exe</code> . ^[13]
		.012	Process Hollowing	TrickBot injects into the svchost.exe process. ^{[1][5][6][8]}
Enterprise	T1090	.002	Proxy: External Proxy	TrickBot has been known to reach a command and control server via one of nine proxy IP addresses. ^[15] ^[11]
Enterprise	T1219		Remote Access Tools	TrickBot uses vncDll module to remote control the victim machine. ^{[9][10]}
Enterprise	T1021	.005	Remote Services: VNC	TrickBot has used a VNC module to monitor the victim and collect information to pivot to valuable systems on the network ^{[18][11]}
Enterprise	T1018		Remote System Discovery	TrickBot can enumerate computers and network devices. ^[8]
Enterprise	T1053	.005	Scheduled Task/Job: Scheduled Task	TrickBot creates a scheduled task on the system that provides persistence. ^{[1][5][6]}
Enterprise	T1553	.002	Subvert Trust Controls: Code Signing	TrickBot has come with a signed downloader component. ^[8]
Enterprise	T1082		System Information Discovery	TrickBot gathers the OS version, machine name, CPU type, amount of RAM available, and UEFI/BIOS firmware information from the victim's machine. ^{[1][2][8][14]}
Enterprise	T1016		System Network Configuration Discovery	TrickBot obtains the IP address, location, and other relevant network information from the victim's machine. ^{[1][7][8]}
Enterprise	T1033		System Owner/User Discovery	TrickBot can identify the user and groups the user belongs to on a compromised host. ^[8]
Enterprise	T1007		System Service Discovery	TrickBot collects a list of install programs and services on the system's machine. ^[1]
Enterprise	T1552	.001	Unsecured Credentials: Credentials In Files	TrickBot can obtain passwords stored in files from several applications such as Outlook, Filezilla, OpenSSH, OpenVPN and WinSCP. ^{[7][8]} Additionally, it searches for the ".vnc.link" affix to steal VNC credentials. ^[12]
		.002	Unsecured Credentials: Credentials in Registry	TrickBot has retrieved PuTTY credentials by querying the <code>Software\SimonTatham\PuTTY\Sessions</code> registry key ^[12]
Enterprise	T1204	.002	User Execution: Malicious File	TrickBot has attempted to get users to launch malicious documents to deliver its payload. ^{[12][8]}
Enterprise	T1497	.003	Virtualization/Sandbox Evasion: Time Based Checks	TrickBot has used <code>printf</code> and file I/O loops to delay process execution as part of API hammering. ^[13]

Groups That Use This Software

ID	Name	References
G0102	Wizard Spider	^{[19][20][21][4][22][23]}
G0092	TA505	^{[24][25]}

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